

HepatoPattern

Machine-Learning Classification of Liver Cirrhosis & Disease Progression from Biochemical Biomarkers

Disease Pattern Classification

PROJECT DOCUMENTATION & REPORT

615

patient records

12

input features

97.6%

test accuracy

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[Notebook Link](#)



[GitHub Link](#)



[Streamlit App Link](#)



[Dataset Link](#)

Summary & Contents

Executive summary

HepatoPattern is a supervised binary classifier that separates active liver pathology (hepatitis, fibrosis, cirrhosis) from healthy blood donors using 12 demographic and biochemical features from the Hannover Medical School HCV cohort (615 patients). A leakage-free pipeline (median imputation, standard scaling, classifier) was benchmarked with 5x3 repeated stratified cross-validation across Logistic Regression, Decision Tree and Random Forest. Random Forest won (CV accuracy 95.9%, ROC-AUC 0.978) and scored 97.6% accuracy, ROC-AUC 0.997 and 100% precision on a held-out test set of 123 patients. A 6-marker routine panel keeps 95.9% accuracy. The solution is delivered as a Streamlit application.

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1. Problem Definition



“A healthcare dataset contains observations associated with different health conditions. Develop a system to investigate and classify the relevant outcomes.”

1.1 Problem formulation

Item	Decision
Clinical domain	Liver cirrhosis & disease progression (chronic Hepatitis C: hepatitis, fibrosis, cirrhosis)
ML task	Supervised binary classification (clinical risk screening)
Inputs	Age, Sex and 10 routine blood biomarkers (liver enzymes, synthesis and metabolic markers)
Output	Class 1: Active Liver Pathology Class 0: Healthy Blood Donor, with a risk probability
Target mapping	1 = Hepatitis, Fibrosis, Cirrhosis (75) 0 = Blood Donor, suspect Blood Donor (540)
Metrics	Accuracy, ROC-AUC, Recall, Precision, F1-score, Confusion Matrix
Project title	HepatoPattern: ML Classification of Liver Cirrhosis & Disease Progression from Biochemical Biomarkers

1.2 Why error costs are asymmetric



False negative: HIGH cost

Active hepatitis or fibrosis is missed, so it can progress silently to irreversible cirrhosis and liver failure, with a risk of viral transmission.



False positive: LOW cost

A healthy person is flagged, which prompts a repeat blood test and viral RNA confirmation. Clinical harm is minimal.

1.3 Why accuracy alone is not enough

Only 12.2% of patients are positive, so a model that always predicts “healthy” already scores 87.8% accuracy. Accuracy is therefore reported together with recall (missed cases), precision (false alarms), F1, ROC-AUC and the confusion matrix, which together show how well the minority class is actually detected.

2. Dataset & Documentation

The data is the Hannover Medical School Hepatitis C & Cirrhosis cohort, a clinical laboratory benchmark collected by the Medical Informatics Institute and published in the UCI Machine Learning Repository (Dataset ID #571, HCV data). The notebook downloads and extracts the archive automatically from archive.ics.uci.edu and saves it as `cirrhosis.csv`, so the work is fully reproducible. The dataset was chosen because it holds real laboratory values for patients across the full hepatitis-to-cirrhosis spectrum, plus healthy donors as controls.

615

patient records

12

input features (+ category label)

540 : 75

healthy : pathology

2.1 Variable dictionary

Variable	Description	Normal range	Group
Age	Patient age (19 to 77 years)	—	Demographic
Sex	Female / Male (coded f=0, m=1)	—	Demographic
ALT	Alanine aminotransferase (SGPT)	10–50 U/L	Liver enzyme
AST	Aspartate aminotransferase (SGOT)	10–45 U/L	Liver enzyme
GGT	Gamma-glutamyl transferase	10–60 U/L	Liver enzyme
ALP	Alkaline phosphatase	40–130 U/L	Liver enzyme
ALB	Serum albumin	35–52 g/L	Synthesis
BIL	Total bilirubin	3–21 µmol/L	Synthesis
CHE	Cholinesterase (falls in hepatic failure)	4.5–12.0 kU/L	Synthesis
CHOL	Cholesterol	—	Metabolic
CREA	Creatinine	—	Metabolic
PROT	Total protein	—	Metabolic

Category labels in the source file: 0 = Blood Donor, 0s = suspect Blood Donor, 1 = Hepatitis, 2 = Fibrosis, 3 = Cirrhosis. Dashes mark variables for which no reference range is given in the project documentation.

3. Data Quality & Preprocessing

3.1 Data-quality observations

Feature	Missing values	Share of 615
ALP	18	2.9%
CHOL	10	1.6%
ALB, ALT, PROT	1 each	0.2% each
All other features	0	0%

The file loads as 615 rows x 13 columns (12 features plus Category). Only 5 features have gaps, with 31 missing cells in total. The median is used for imputation because enzyme values are right-skewed with outliers.

3.2 Preprocessing steps

1

Target encoding: Category containing “Blood Donor” becomes 0, everything else (Hepatitis, Fibrosis, Cirrhosis) becomes 1.

2

Sex encoding: Sex is mapped to a numeric code: m = 1, f = 0.

3

Feature sets: Full panel of 12 features, and a routine panel of 6 (Age, Sex, ALB, ALT, AST, BIL).

4

Stratified split: 80/20, random_state = 42: 492 training (60 pathology, 432 donors) and 123 test (15 pathology, 108 donors).

5

Median imputation: SimpleImputer(strategy = median), fitted on training data only.

6

Standard scaling: StandardScaler puts U/L, g/L and $\mu\text{mol/L}$ features on one scale.

3.3 Leakage-free pipeline



All fitted steps live inside one scikit-learn Pipeline, so cross-validation folds and the test set never influence imputation or scaling.

4. Exploratory Data Analysis

4.1 Class distribution

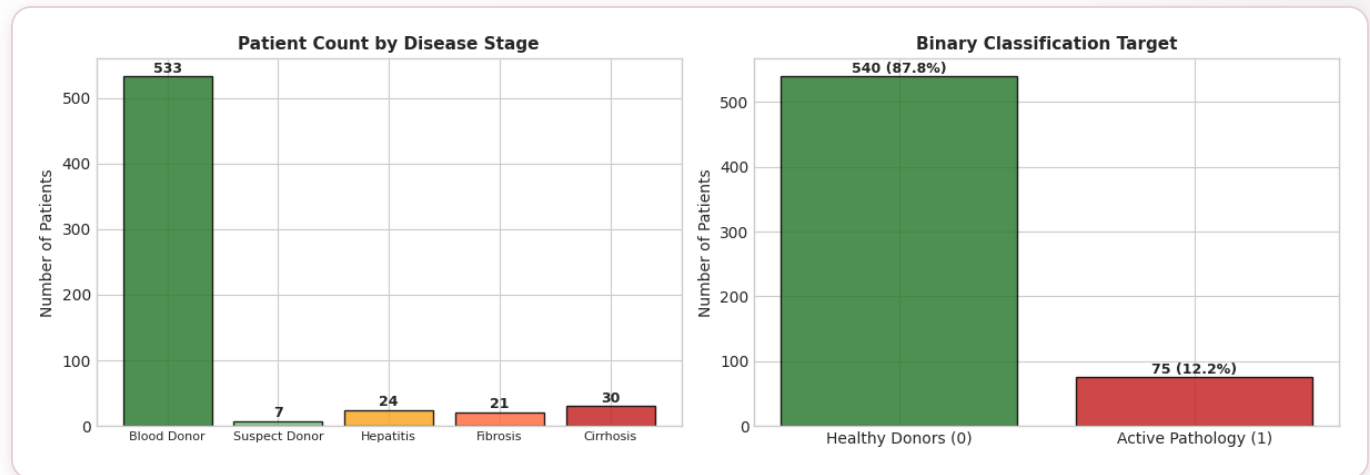


Figure 1: Patient count by disease stage (left) and the binary target (right)

- Strong imbalance: 540 healthy (87.8%) vs 75 pathology (12.2%), about 7 to 1.
- Pathology is spread over three stages: hepatitis 24, fibrosis 21, cirrhosis 30.
- Consequence: stratified splitting and metrics beyond accuracy are required.

4.2 Key biomarker comparison

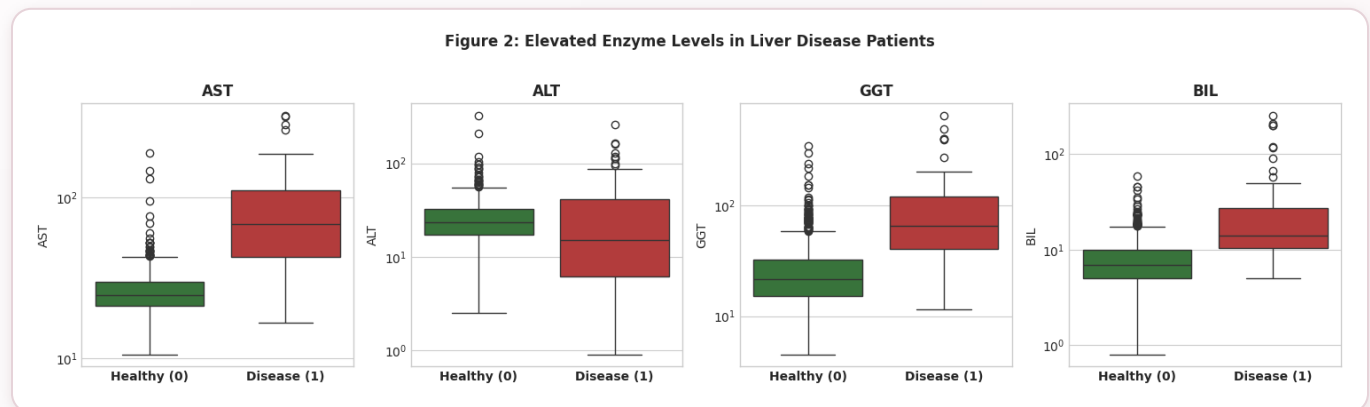


Figure 2: AST, ALT, GGT and bilirubin in healthy vs pathology patients (log scale)

- AST, GGT and bilirubin are clearly higher in pathology patients.
- ALT shows no clear elevation: its median is not higher in the pathology group, and the ranges overlap heavily.
- No single marker separates the classes, which motivates a multivariate model.

5. Model Development & Comparison

5.1 Models compared

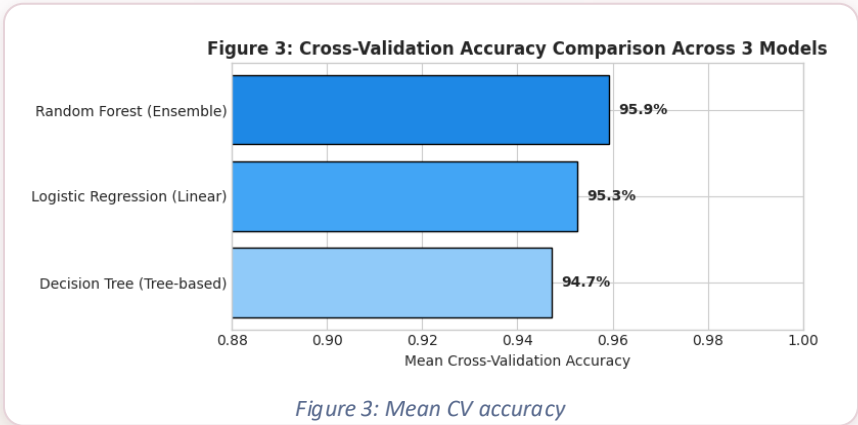
Model	Paradigm	Configuration	Why chosen
Logistic Regression	Linear baseline	max_iter = 1000	Simple, interpretable benchmark. Assumes a linear boundary.
Decision Tree	Tree-based	max_depth = 4	Learns non-linear, rule-like splits. High variance.
Random Forest	Ensemble	200 trees, max_depth = 6	Bagged, de-correlated trees capture interactions and cut variance.

5.2 Evaluation protocol

Every model sits in the same pipeline (median imputer, standard scaler, classifier) and is scored on the 492-patient training set with Repeated Stratified 5-Fold cross-validation, 3 repeats (15 folds), random_state = 42. The 123-patient test set is not used for model selection.

5.3 Cross-validation results

Model	CV Accuracy	CV F1	CV Recall	CV ROC-AUC
Random Forest (Ensemble)	0.959	0.813	0.739	0.978
Logistic Regression (Linear)	0.953	0.771	0.700	0.968
Decision Tree (Tree-based)	0.947	0.767	0.717	0.836



- Random Forest is best on all four metrics.
- Its edge in recall and F1 shows better minority-class detection.
- Decision Tree AUC (0.836) is lowest: coarse probabilities.

6. Evaluation on the Held-Out Test Set

6.1 Headline results (123 patients: 15 pathology, 108 donors)

97.6%

Accuracy

0.997

ROC-AUC

100%

Precision

80.0%

Recall

100%

Specificity

6.2 Classification report

Class	Precision	Recall	F1-score	Support
Healthy Donors (0)	0.97	1.00	0.99	108
Liver Disease (1)	1.00	0.80	0.89	15
Macro average	0.99	0.90	0.94	123
Weighted average	0.98	0.98	0.97	123

6.3 Confusion matrix and ROC curve

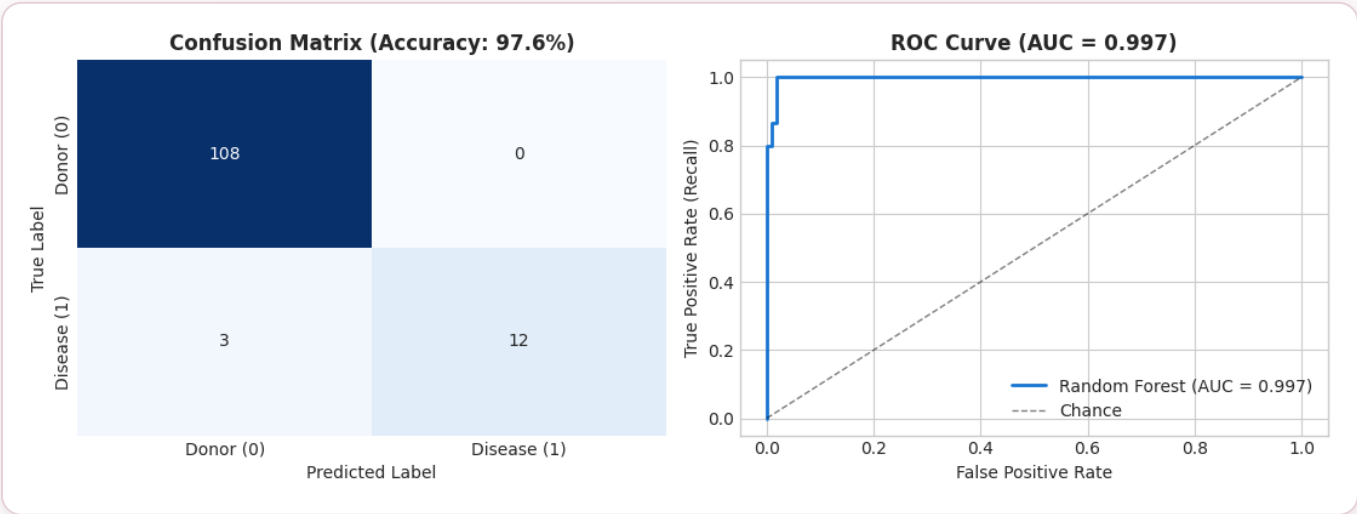


Figure 4: Confusion matrix (accuracy 97.6%) and ROC curve (AUC = 0.997)

6.4 Interpretation

- 120 of 123 patients are classified correctly: 108 true negatives, 12 true positives.
- No false alarms (precision and specificity 100%).
- All 3 errors are false negatives, so recall is 80% (12 of 15 cases found).
- ROC-AUC 0.997 means the risk ranking is almost perfect even where the 0.5 cut-off misses cases.

7. Feature Study & Interpretability

7.1 Full clinical panel vs routine outpatient panel

Feature panel	Features	Test accuracy	Test ROC-AUC
Full clinical panel (12)	All 12 features	97.56%	0.9969
Routine outpatient LFT (6)	Age, Sex, ALB, ALT, AST, BIL	95.93%	0.9914

Dropping six specialised tests (CHE, CHOL, CREA, GGT, ALP, PROT) costs only about 1.6 points of accuracy and 0.006 of AUC. A basic 6-marker blood panel is therefore enough for accurate liver triage, which supports use in primary-care clinics.

7.2 Feature importance (Random Forest, Gini)

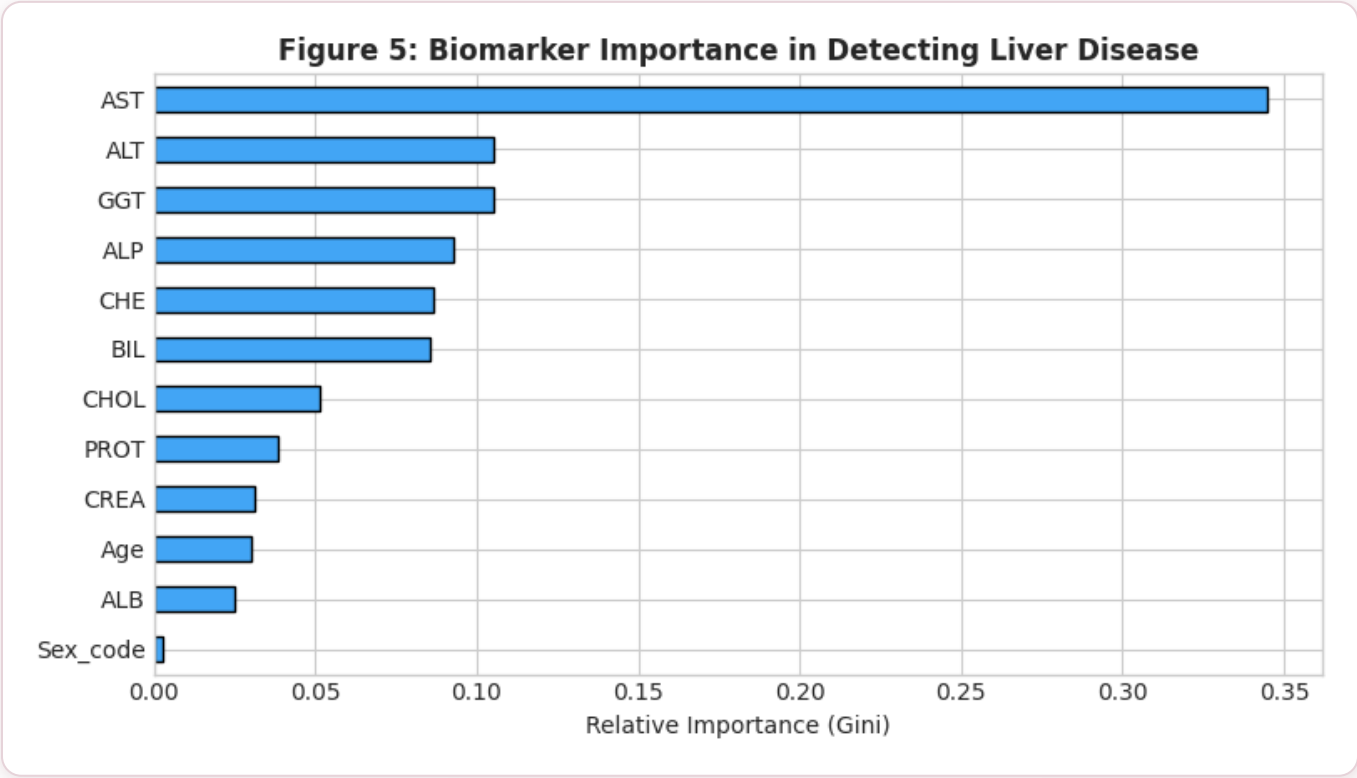


Figure 5: Relative biomarker importance in the full-panel Random Forest

- AST is the dominant predictor (about 0.35), well ahead of the rest.
- ALT and GGT follow (about 0.10 each), then ALP, CHE and bilirubin.
- Albumin, age and especially sex carry little weight in this model.
- The ranking is clinically plausible: hepatocyte injury enzymes lead, followed by synthetic and excretory markers. Gini importance can favour high-variance features, so it is read as a guide, not proof of causation.

8. Error Analysis & Limitations



Error analysis

- 3 errors in 123, all false negatives (pathology predicted healthy).
- 0 false positives at the default 0.5 threshold.
- With only 15 positives, one missed case moves recall by 6.7 points, so the test estimate is uncertain.
- Overlap in ALT and other markers means some patients look healthy on blood tests alone.
- A lower decision threshold trades extra false alarms for fewer missed cases; the app offers a threshold slider.



Limitations

- Pathology is mainly chronic Hepatitis C, so other causes (alcohol, NAFLD / NASH) are not covered.
- Retrospective single-source data: 615 patients, only 75 positive.
- Controls are blood donors, who are healthier than typical clinic patients, which can make the task look easier.
- Binary target hides stage differences between hepatitis, fibrosis and cirrhosis.

8.1 Future work



External validation

Test on NAFLD / NASH and other cohorts.



Prospective study

Multi-centre validation at the bedside.



Multimodal fusion

Add FibroScan stiffness (kPa) to blood tests.

Ethical note

HepatoPattern is an educational and research prototype. It is not a certified medical device and must not be used for real clinical diagnosis or treatment planning. Any positive result should be confirmed by a clinician with repeat testing.

9. Streamlit Application

The trained Random Forest pipelines are saved as joblib artifacts (full and routine panel, plus a meta.json with the project summary) and served through main.py, a Streamlit app with two modes selected from the sidebar.

9.1 Data flow



Simple Nurse Patient Intake Form

- For bedside nursing staff and outpatient triage, under 1 minute
- Cards for demographics, routine labs and optional advanced tests
- Quick-fill presets: healthy blood donor and active cirrhosis patient
- Plain-English output with key abnormal values and next steps
- Model badge: 97.6% test accuracy | 0.997 ROC-AUC



Research & Benchmark Dashboard

- Sidebar: choose the 12-feature or 6-feature panel
- Decision-threshold slider to explore sensitivity trade-offs
- Tabs: Predict Patient Outcome (with PCA cohort view), Data Explorer, Model Performance, About
- Live confusion matrix, ROC curve and feature importances

Run: `uv run streamlit run main.py` (opens <http://localhost:8501>)

10. Repository & Reproducibility

10.1 Project structure

```
.
├── artifacts
│   ├── meta.json
│   ├── rf_model_full.joblib
│   └── rf_model_routine.joblib
├── cirrhosis.csv
├── main.py
├── problem_statement.txt
├── pyproject.toml
├── README.md
├── requirements.txt
├── RohanVashisht150096724132.ipynb
└── uv.lock
```

10.2 File roles

File	Purpose
RohanVashisht150096724132.ipynb	Full research notebook: EDA, preprocessing, models, evaluation
main.py	Streamlit application (nurse form + research dashboard)
cirrhosis.csv	Hannover HCV and cirrhosis cohort (UCI #571)
artifacts/	Saved Random Forest pipelines (full, routine) and meta.json
problem_statement.txt	Case-study prompt, objectives and deliverables
pyproject.toml, uv.lock	uv project configuration and locked dependencies
requirements.txt	pip fallback dependency list
README.md	Project documentation and clinical summary

10.3 How to run

```
uv sync                                # create .venv, install deps
uv run streamlit run main.py           # launch the app
uv run jupyter notebook RohanVashisht150096724132.ipynb
pip install -r requirements.txt        # fallback without uv
```

11. Conclusion & Deliverables Mapping

Conclusions

- A clean, leakage-free pipeline classifies liver pathology from routine blood tests.
- Random Forest beat Logistic Regression and Decision Tree in cross-validation (accuracy 95.9%, AUC 0.978).
- Held-out test: 97.6% accuracy, 0.997 ROC-AUC, 100% precision, 80% recall.
- A 6-marker routine panel retains 95.9% test accuracy, suited to basic clinics.
- AST, ALT, GGT, ALP, CHE and bilirubin drive the predictions.

11.1 Deliverables mapping

Required deliverable	Report page	Source location
Problem definition & title	3	Notebook Section 1, README
Dataset & documentation	4–5	Notebook Section 2, cirrhosis.csv
Exploratory data analysis	6	Notebook Section 3
Preprocessing	5	Notebook Section 4
Model development & comparison	7, 9	Notebook Sections 5 and 6
Evaluation, errors, limitations	8, 10	Notebook Sections 7 to 9
Streamlit application	11	main.py, artifacts/
Report, code, demo, presentation	all	This report, source code, viva deck

11.2 References

- UCI Machine Learning Repository: HCV data, Dataset ID 571. archive.ics.uci.edu/dataset/571/hcv+data
- Lichtinghagen R. et al. The Enhanced Liver Fibrosis (ELF) score: normal values, influence factors and proposed cut-off values. *J Hepatol* 2013;59:236–42.
- Hoffmann G. et al. Using machine learning techniques to generate laboratory diagnostic pathways: a case study. *J Lab Precis Med* 2018;3:58.
- Pedregosa F. et al. Scikit-learn: Machine Learning in Python. *JMLR* 12, 2011.